

Play Before Pixels · Research hub

Research hub review bundle

Six plain-language pages about the term “virtual autism”, screens and early development, prepared for three paid reviews: an autistic sensitivity read, a clinician review of the practical guidance, and a claims and legal review.

VERSION ID	BUILT	STAGE	WORDS / BLOCKS
0e756e88c254	2026-09-28	Pre-verification draft	11,942 / 332

Pre-verification draft. The sources on these pages have not yet been checked against the original papers, so numbers and items marked [VERIFY] may still change. Use this copy to scope and quote the work. The review itself is done on the *review version*, rebuilt after verification, with a new version ID. Items still marked [VERIFY] in this build: **184**.

Confidential. These pages are unpublished drafts owned by AlphaPlay LLC (doing business as Play Before Pixels). Please do not share them, quote them, or upload them to any AI tool or public service, and do not contact any person or organization named in them. Your contract sets out the full terms.

Pages and who reviews what

Code	Page and planned web address	Words	A. Autistic sensitivity read	B. Clinician review	C. Claims and legal
IDX	What is "virtual autism"? What the research says (and doesn't) /research/virtual-autism/	3,794	Full read; focus: opening boxes; A note to autistic readers and their families; What critics say; What parents can do	Opening boxes; What we still don't know; What parents can do (880 words)	Whole page
FAQ	Questions parents ask about "virtual autism" and screens /research/virtual-autism/faq/	2,574	Full read; focus: About the term; For autistic readers	Opening boxes; If you're worried; Everyday life with screens (1,031 words)	Whole page
GLO	Glossary /research/glossary/	2,272	Full read; focus: Words about the debate; Words about autism and development	Words about autism and development; Tests and checklists; Help and services (877 words)	Whole page
PED	Questions to ask your pediatrician /research/pediatrician-questions/	593	Full read; focus: About us as a family	Whole page (593 words)	Whole page
EIV	How to ask for a free early intervention evaluation /research/early-intervention/	1,450	Full read; focus: Common worries	Whole page (1,450 words)	Whole page
POL	Editorial policy /research/editorial-policy/	1,259	Full read; focus: Respectful language	—	Whole page
Total		11,942	11,942 words (3,658 in focus sections)	4,831 words in scope	11,942 words

How to use this bundle

Each page starts on a new sheet. Read the pages your brief covers, write your comments in the comment table at the back (or the same columns in any spreadsheet), then complete your sign-off form. All communication is in writing.

IDX-014

Block ID. Every paragraph, list item and box has an ID in the left margin: page code plus number. Use it in your comments so each one points to one exact place.

CLINICIAN SCOPE

Clinician scope. Sections in the clinician's scope carry this tag, and each of their blocks has a blue bar in the margin. Pages marked "whole page" in the table above are entirely in scope.

SENSITIVITY FOCUS

Sensitivity focus. The sensitivity reader reads every page; these sections are where respect and framing matter most.

[VERIFY]

Not yet checked. The editors have not yet read this source or official page themselves. On the review version, every remaining claim has been checked or removed.

[5]

Placeholder. A number or promise the business still has to confirm. Tell us if the wording around it is a problem.

Underlined text

A link on the website, usually to a one-page summary of a study. Study pages are not part of this bundle.

What every reviewer should know

- The business behind these pages sells play products (books and printables). The research pages carry no product links and no prices; one link goes to a free play printable, described only as play and family time. You are not asked to review, and must not be presented as endorsing, any product.
- The pages were drafted with the help of AI tools and edited by the business. You are reviewing them, not writing them.
- Your fee is the same whatever you conclude. “Does not pass” is a valid result.
- Your name appears on a page only if you choose that on your sign-off form, only on the pages you reviewed, and only for the version you reviewed.
- The internal editor notes in the source files are not shown here. Each page ends with the list of sources recorded for it.

Comment levels

- **Must-fix:** the page cannot be published with this as written.
- **Should-fix:** strongly recommended; the business must answer in writing if it does not change it.
- **Suggestion:** optional improvement.

What is "virtual autism"? What the research says (and doesn't)

Draft status: draft: secondary sources only; not cleared to publish until every cited source is read (abstract at minimum) and the review steps in editorial-policy.md are done · **Last reviewed:** 2026-09-28 · **Words:** 3,794 (clinician scope 880) · **Source file:** content/research-hub/index.md · **SHA-256:** b67947fe4d0e

IDX-001 "Virtual autism" is a term some clinicians use for autism-like behaviors seen in some young children with heavy early screen exposure. It is not a medical diagnosis. Studies have found links (associations) between early screen time and later outcomes, not proof that screens cause autism, and autism is not a parent's fault. If you're worried about your child, talk with your pediatrician and ask about a free early intervention evaluation.

IDX-002 **Worried? You don't need to wait.**

1. Talk with your pediatrician or family doctor.
2. Ask for a free evaluation. In the US you can ask your state's early intervention program yourself. [How to ask.](#)
3. Bring [these questions.](#)

This is not medical advice. This page explains research. It can't tell you whether your child is autistic, has a delay, or is developing as expected. Only a professional who meets your child can.

IDX-003 **Last reviewed:** [set to the day all four review gates pass; never the drafting date]

The short version

- IDX-004 • **It is not a diagnosis.** "Virtual autism" is not in the World Health Organization's ICD or the DSM, and even the psychologist who coined it says it is not a medical diagnosis ([Zamfir 2018](#) [\[VERIFY\]](#)).
- IDX-005 • **Where it comes from.** The evidence that uses the label is case reports, small clinic series without comparison groups, and surveys at one point in time. We have found no randomized trials, and no long-term studies designed to test whether signs change when screens are reduced ([Ozyazici 2026](#) [\[VERIFY\]](#)).
- IDX-006 • **Larger studies find links, not causes, and not always the same link.** Some large studies find that more early screen time goes with somewhat higher scores on autism screening checklists ([Heffler 2020](#)), or with higher odds of an autism diagnosis in some groups ([Kushima 2022](#), boys only). Another large study found no simple "more screens, more autism" pattern ([Melchior 2022](#) [\[VERIFY\]](#)).
- IDX-007 • **But the links may run the other way, or come from something else.** Children with a higher genetic likelihood of autism already watch more as toddlers ([Takahashi N 2023](#) [\[VERIFY\]](#)). The largest review we have found reported an overall positive link, but it shrank and stopped being statistically significant once the authors corrected for publication bias ([Ophir 2023](#) [\[VERIFY\]](#)). Two further studies built to get closer to cause found no clear causal signal, though each has important limits: one rests on a debatable statistical assumption, and the other cannot test screen use in infancy ([Lin 2025](#) [\[VERIFY\]](#); [Cai 2025](#) [\[VERIFY\]](#)). None of this rules out that heavy screen use plays some part for some children. It means that has not been shown.
- IDX-008 • **Researchers, clinicians and autism organizations have objected to the term.** They say it risks bringing back the old, false idea that parents cause autism, and that it blurs what autism is ([Krijnen 2026](#) [\[VERIFY\]](#); [autismus Deutschland](#) [\[VERIFY\]](#)).

IDX-009 • **What helps right now does not depend on settling the debate.** If you're worried, ask your pediatrician and ask for a free evaluation. Talking, playing and reading together are good for every child, whatever the research eventually shows.

IDX-010 **How we wrote this:** every claim links to a study page that says exactly what we read. Sources marked **[VERIFY]** are ones we have not yet been able to check against the original paper. Read them as provisional. Our [editorial policy](#) explains how we choose sources, who reviews them, and how to report a mistake.

A note to autistic readers and their families SENSITIVITY FOCUS

IDX-011 If you are autistic, or you love someone who is, parts of this page may be hard to read. Much of the research below treats autism as something to be counted and explained, and the term "virtual autism" has been criticized for the hurt it can cause. Here is where we stand.

IDX-012 • **Autism is a lifelong neurodevelopmental difference, not damage and not a failure.** Autistic children and adults are whole people. Nothing on this page is meant to suggest that an autistic child is a bad outcome, or that autism is something to be prevented or fixed.

IDX-013 • **Support needs differ.** Some autistic children need a lot of support with communication, daily life, sleep or health. Others need much less. Both are real, and every family's experience deserves respect.

IDX-014 • **Autism is not caused by parenting.** The old idea that parents cause autism was wrong, and autism organizations and researchers warn that "virtual autism" risks bringing it back ([autismus Deutschland](#) **[VERIFY]**; [Krijnen 2026](#) **[VERIFY]**). No study has shown that screens cause autism.

IDX-015 • **Screens can meet real needs.** An autistic researcher **[VERIFY from the paper itself that the author identifies as autistic; if not confirmed, write "A researcher" and drop "their own"]** describes how screen use met their own needs for predictability, safety, autonomy and belonging, and argues that research should ask how and why autistic people use media and weigh benefits as well as risks ([van Asselt 2026](#) **[VERIFY]**).

IDX-016 • **A communication device is a voice, not "screen time".** Tablets and phones used for AAC (augmentative and alternative communication) are communication tools. If your child uses one, keep it within reach.

IDX-017 • **Research words are not our words for people.** Studies use clinical terms such as "disorder", "symptoms", "severity" and "risk". We quote them when we describe what a study measured, and we don't use them to describe autistic people. The [glossary](#) explains what they mean.

IDX-018 • **Language.** Many autistic people prefer identity-first language ("autistic child"), and we mostly use it here. Some people and families prefer "child with autism", and that is their choice.

IDX-019 • **Autistic voices belong in this debate.** We give critiques by autistic researchers and statements from autistic-led organizations the same care as the clinical studies, and we are still searching for more of them **[VERIFY: name only authors whose papers say they are autistic]**. Our [editorial policy](#) requires a paid autistic sensitivity read before any research page is published.

IDX-020 • **If your child has just been identified as autistic:** your child is the same child they were yesterday. Support is about what helps your child communicate, feel safe and thrive, not about making them seem less autistic. Autistic adults' own writing and autistic-led organizations are good places to learn what autism is like from the inside.

IDX-021 If anything here is hurtful, wrong or missing, please [tell us](#).

Where the term came from

IDX-022 **Romania, 2018.** Psychologist Marius Teodor Zamfir looked back at the case histories of young children newly diagnosed with autism at two private centres. According to their parents' recollection, many had spent more than four hours a day with screens before age 3. He called this pattern "autism virtual" and

proposed that it resembled childhood autism. The paper appeared in a literary-studies journal, not a medical one. The children it describes **already had autism diagnoses**. [Read our summary](#) [VERIFY].

IDX-023 **France, 2017 to 2019.** In 2017, warnings from a French maternal and child health physician, including a widely shared video, set off public alarm that heavy tablet and phone use could produce an autism-like picture in toddlers ([summary](#) [VERIFY]). In 2018, three French clinicians proposed a different name, "EPEE" (early and excessive screen exposure), which deliberately avoids the word autism ([Marcelli 2018](#) [VERIFY]). In 2019, French child psychiatrist Bruno Harlé published a hypothesis paper arguing that intensive early screen exposure might play a part in autism-like signs in some young children, and proposed the label "virtual autism" ([Harlé 2019](#)). His paper is an argument for an idea, built from case reports. It does not test the idea.

IDX-024 **Earlier roots.** A 2008 US study noticed that rainier counties had higher autism rates and speculated about indoor TV time, but it had no information on any child's screen use ([Waldman 2008](#) [VERIFY]). In 2016 two US researchers proposed an untested model in which heavy early audiovisual stimulation competes with social brain development ([Heffler and Oestreicher 2016](#) [VERIFY]).

IDX-025 **Other labels you may see.** "Digital autism", "electronic screen syndrome" ([Dunckley](#) [VERIFY]), "post-digital nannying autism syndrome" ([Pouretmad 2022](#) [VERIFY]), and in other languages "autisme virtuel", "virtueller Autismus", "autismul virtual" and "sanal otizm". None is a recognized diagnosis. Some critics suggest neutral terms such as "screen-related developmental delay" ([Krijnen 2026](#) [VERIFY]). See the [glossary](#).

What studies have found

IDX-026 It helps to sort the research by **how** it was done, because some designs can answer the question much better than others.

Case reports and small clinic series

IDX-027 A few reports describe young children with heavy screen use and autism-like signs whose developmental scores changed after screen time was cut and adult time added ([Heffler 2022](#), two children [VERIFY]; [an uncontrolled series of 30 children in Pakistan](#) [VERIFY]; [a very small series in Indonesia](#) [VERIFY]).

IDX-028 These reports matter because they are where the idea comes from. But they **cannot tell us why** a child changed. None had a comparison group. In each, less screen time came together with more adult attention and, often, professional support, and young children develop quickly anyway. Assessors usually knew what had changed at home. And a lower score on an autism checklist is not the same as a child being happier or better off.

Surveys at one point in time, and case-control studies

IDX-029 Many studies compare screen use in autistic and non-autistic children, or measure screen time and autism-screening scores at the same time. Autistic toddlers often started screens earlier and watched more ([Chonchaiya 2011](#) [VERIFY]; [Tunisia 2025](#) [VERIFY]). In 5,336 Singapore toddlers, above-average screen time went with about 24% higher odds of a positive autism screen ([Sundarima 2025](#) [VERIFY]). A positive screen is not a diagnosis.

IDX-030 Not every survey finds a difference: in a large US national survey of school-age children, autistic and non-autistic children had similar total screen time ([Montes 2016](#) [VERIFY]).

IDX-031 These designs **cannot tell which came first**. In the Tunisian study, autistic children often preferred repetitive videos and looked at the screen from the side. Those are ways of watching that fit with autism already being there.

Studies that follow children over time

- IDX-032 • **United States, 2,152 children.** TV or video at 12 months went with slightly higher autism-checklist scores at age 2 (4.2% higher), but **not** with screening "at risk". Daily parent-child play went with lower scores (Heffler 2020). Independent experts criticized the crude screen measure and pointed out that no one was diagnosed (Science Media Centre [VERIFY]).
- IDX-033 • **Japan, 84,030 children.** In **boys but not girls**, longer screen time at age 1 went with higher odds of an autism diagnosis by age 3. The odds were higher in each group from 1 hour a day upward, but the estimates for 2 to 4 hours and for 4 hours or more were similar, with widely overlapping ranges, so the study cannot tell whether the odds keep rising above about 2 hours. Autism was recorded only up to age 3 (Kushima 2022).
- IDX-034 • **France, 12,950 two-year-olds.** Screen use went with very slightly higher odds of a *medium* screening score but *lower* odds of a *high-risk* score. There was no simple "more screens, more autism" pattern (Melchior 2022 [VERIFY]).
- IDX-035 • **Australia, 5,107 children.** More than 14 hours a week at age 2 went with autism by 12 in a standard analysis, but not in a second analysis designed to get closer to cause, which itself rests on a debatable assumption (Lin 2025 [VERIFY]).
- IDX-036 • **United States, 82 toddlers, many with a family history of autism or ADHD.** Children later identified with autism or ADHD concerns had more screen time at 18 months. The study cannot tell whether that screen time was an early sign or a contributor (Hill 2024 [VERIFY]).

Reviews that combine many studies

- IDX-037 • The **largest** review we have found pooled 46 studies with 562,131 people. It found a positive link overall, but also strong signs of **publication bias**: studies with positive findings were more likely to be published. After a standard correction, the link was much smaller and **no longer statistically significant**. The authors concluded that the proclaimed association is "not sufficiently supported" (Ophir 2023 [VERIFY]).
- IDX-038 • Two other reviews report a link (Yuan [VERIFY]; Liu 2025 [VERIFY]). Whether they corrected for publication bias is not yet confirmed. Another found the evidence mixed and too thin to confirm any link (Sarfraz 2023 [VERIFY]).

Studies that test other explanations

- IDX-039 • **Genes.** In Japan, children with a higher genetic likelihood of autism already spent more time on screens as toddlers. The authors suggest long screen time may be an **early sign** rather than a cause (Takahashi N 2023 [VERIFY]). In older US children, shared genetic risk explained part of the link between screen time and attention and emotional problems (Zhang 2023 [VERIFY]).
- IDX-040 • **Natural experiments with genes.** A Mendelian randomization study found no evidence that genetically predicted leisure screen time affects autism. Its genetic markers reflect mostly adult screen habits, so it cannot test infant exposure directly (Cai 2025 [VERIFY]).

Screens and talk: related research that is **not** about autism

- IDX-041 • A separate body of research looks at screens and everyday talk and development. It is often mixed up with the autism question, so we keep it apart:
- IDX-042 • More screen time at age 1 went with higher odds of communication and problem-solving delays on a screening questionnaire at 2 and 4 (Takahashi I 2023, 7,097 children).
- IDX-043 • In home audio recordings, screen time at 36 months went with less talk. Scaled to the average amount of daily screen time in the study, that was 1,139 fewer adult words, 843 fewer child vocalizations and 194 fewer back-and-forth conversational turns a day (Brushe 2024).

- IDX-044 • Background TV went with weaker language, and watching *together* went with stronger language (Madigan 2020 [VERIFY]; Mallawaarachchi 2024 [VERIFY]).
- IDX-045 These are associations too. They are **not autism studies**, and we never present them as evidence about autism.

What critics say SENSITIVITY FOCUS

- IDX-046 Criticism of the term started almost as soon as the term did:
- IDX-047 • **Romania.** A Romanian psychology site called the original study an essay lacking scientific rigor, not published in a recognized scientific journal (summary [VERIFY]).
- IDX-048 • **France, 2017-18.** French commentators called the viral alert subjective and without scientific foundation (summary [VERIFY]).
- IDX-049 • **UK, 2020.** Independent researchers said the 2020 US study's screen measure was crude and that no child was diagnosed (Science Media Centre [VERIFY]). A media and disability researcher published a letter calling for better research (Alper 2020 [VERIFY]).
- IDX-050 • **Germany.** The national autism association says the term is not a recognized medical term and warns that it revives the false idea that autism comes from parenting (autismus Deutschland [VERIFY]).
- IDX-051 • **Journals, 2024 onward.** Reviews and commentaries from India, the Netherlands and Nepal conclude that the evidence is too preliminary for a diagnosis, that the link may run both ways, and that the label adds stigma for families of autistic children (Detroja and Bhatia 2024 [VERIFY]; Krijnen 2026 [VERIFY]; Dhungel 2026 [VERIFY]). An autistic researcher [VERIFY identity from the paper itself] argues that research should ask how and why autistic people use media, and weigh benefits as well as risks (van Asselt 2026 [VERIFY]).
- IDX-052 Their shared points: the term is not a diagnosis; it risks bringing back parent blame; it blurs what autism is; and some replacement terms (such as "screen-induced developmental deviation") still assume a cause that has not been shown.

Why a link is not proof of cause

- IDX-053 Almost every study above is **observational**: researchers measured what families were already doing. That can show that two things go together. It cannot, on its own, show that one produces the other. Here is why, in plain terms.
- IDX-054 **1. Reverse causation: the arrow may point the other way.** Some babies and toddlers who will later be diagnosed as autistic may find screens especially calming, predictable or interesting, well before anyone notices other signs. Parents may also, understandably, use screens more with a child who connects with people in different ways or is hard to settle. In either case, *early autistic traits would lead to more screen time*, rather than the other way round. The Japanese genetic study is one piece of evidence that this can happen (Takahashi N 2023 [VERIFY]). A large Japanese cohort on general development (not autism) also found the link running both ways: lower developmental scores went with more screen time later in some families (Yamamoto 2023 [VERIFY]).
- IDX-055 **2. Confounding: a third factor behind both.** Imagine a family under strain: long work hours, no extended family nearby, a parent who is unwell. That situation might mean more screen time *and* fewer chances for back-and-forth play, and it might also affect when a child gets assessed. If a study doesn't measure that strain, screens can look responsible for something the strain is really driving. Genes can be a confounder too, because autism runs in families, and parents' own traits can shape how a household uses screens.

- IDX-056 **3. Memory and measurement.** Many studies ask parents to remember how much their child watched years earlier, often *after* a diagnosis. Worried parents may remember differently. Most studies also use **screening checklists**, which flag children for a closer look. A high checklist score is not a diagnosis.
- IDX-057 **4. Publication bias.** Studies that find a striking link are more likely to be published and shared than studies that find nothing. That can make a link look stronger than it is. This is what the largest review we have found reported ([Ophir 2023](#) **[VERIFY]**).
- IDX-058 **5. Change over time happens anyway.** Young children, including autistic children, develop and change quickly. When a report says a child "improved" after screens were cut, we can't tell whether that was the screens, the extra adult time that replaced them, other support, or simply growing up, unless there is a comparison group.
- IDX-059 **What would show cause?** A randomized trial (families randomly assigned to change screen habits or not, with assessors who don't know which), or long-term studies that measure children carefully before and after, with other explanations accounted for. As of this review, we have not found a randomized trial of screen reduction and autism-like signs in young children ([Ozyazici 2026](#) **[VERIFY]**). The closest is a small Iranian study that combined a parent-child interaction programme with screen reduction; it changed two things at once, and its design is not yet confirmed ([Sadeghi 2021](#) **[VERIFY]**).

What we still don't know **CLINICIAN SCOPE**

- IDX-060 • Whether heavy early screen time plays **any** causal part in autism-like signs, and if so, for which children.
- IDX-061 • Whether some very young children with heavy screen use show difficulties that look like autism but are something else, and how clinicians should tell the difference. Only a qualified clinician assessing a child over time can judge that.
- IDX-062 • Whether *when* screens are used (for example, before sleep) or *how* (alone or together, background or foreground) matters more than total time ([Lin YH 2022 letter](#) **[VERIFY]**; [Tbilisi 2025](#) **[VERIFY]**).

What parents can do **CLINICIAN SCOPE** **SENSITIVITY FOCUS**

- IDX-063 **1. If something worries you, ask. You don't need to wait.** Talk with your pediatrician or family doctor about any concern: how your child talks, plays, responds, moves or connects with you. Bring notes and examples. Our [printable list of questions for your pediatrician](#) can help. If your child has stopped using words or skills they used to have, tell your pediatrician soon rather than waiting for the next routine visit **[VERIFY wording against CDC "Learn the Signs. Act Early." and AAP guidance; clinician reviewer to confirm]**.
- IDX-064 **2. In the US, you can ask for a free early intervention evaluation yourself.** Every US state has an early intervention program for children under 3 under federal law (IDEA Part C). You can usually contact it directly without a doctor's referral, and the evaluation is free **[VERIFY against official program pages]**. For children 3 and older, contact your local public school district and ask for a free evaluation **[VERIFY]**. [How to ask, step by step](#), with pointers for the UK, Canada and Australia.
- IDX-065 **3. You don't have to choose between "screens" and "evaluation".** Some families decide to change screen habits while they wait. That's fine, and if you notice changes, tell the people evaluating your child: what you see at home is useful to them. But please don't wait to see whether cutting screens makes a difference before asking for an evaluation. Support is worth having whatever the reasons behind what you're noticing, and an evaluation is how you find out what your child needs.
- IDX-066 **4. Everyday play and conversation, as ordinary family life.** Talking, singing, reading and playing together are good for every child, autistic or not. They are not a treatment, they are not a way to make a child less autistic, and they are not a test you can fail. A few easy ways in:

- IDX-067 • **Pause and wait.** Leave a little gap so your child can take a turn, with a sound, a look, a gesture or a word.
- IDX-068 • **Say what you see.** "Big ball. Rolling. Stop!"
- IDX-069 • **Repeat and add one word.** Child: "Car." You: "Blue car."
- IDX-070 • **Offer a choice.** "Banana or apple?" A point, a reach or a look counts as an answer.
- IDX-071 • **Follow their lead.** Join in with what your child is already interested in, including lining things up, spinning wheels or watching the same thing again. Their interests are a way in, not a problem.
- IDX-072 • **Sing and gesture.** Songs with actions invite turn-taking without pressure.
- IDX-073 • **Use your home language.** Talk, sing and read in the language you know best. Every language counts.
- IDX-074 • **5. What guidelines say about screens for little ones.** The World Health Organization recommends no screen time for babies under 1 and no more than 1 hour a day for 2-to-4-year-olds, with plenty of active play ([WHO 2019](#)). For 1-year-olds, it is recalled as not recommending sedentary screen time at all **[VERIFY against the guideline]**. These limits are precautionary, not a line past which harm has been shown. The American Academy of Pediatrics published detailed advice for under-5s in 2016 ([AAP 2016](#)) and a new policy statement in January 2026 that reportedly replaces its earlier media guidance ([AAP 2026](#) **[VERIFY]**). We will summarize the 2026 advice once we have read it. Research on *how* screens are used suggests that watching together and keeping background TV off may matter as well as total time ([Mallawaarachchi 2024](#) **[VERIFY]**).
- IDX-075 • **6. Go easy on yourself.** Almost every family uses screens. Using a screen so you can shower, cook or catch your breath does not make you a bad parent. No study has shown that screens cause autism. If your child is autistic, you did not cause it, and there is nothing here to feel guilty about. Your child is not a problem to be solved.

All the evidence in one place

- IDX-076 • [The research library](#): every source we use, grouped so that critical and null studies are as easy to find as supportive ones.
 - IDX-077 • [Questions parents ask](#)
 - IDX-078 • [Glossary of research terms](#)
-
- IDX-079 Worried? Talk with your pediatrician. Free early intervention evaluations are available for young children. Asking early is a caring step, and it's not your fault. **[VERIFY access wording against official program pages at publish.]** Outside the US: ask your doctor, nurse or health visitor about free local early support.

SOURCES THIS PAGE LISTS (FROM ITS FRONT MATTER)

1. Harlé B. Trends in Neuroscience and Education. 2019;17:100119. doi:10.1016/j.tine.2019.100119 • status: allowed-list • what we read: secondary-only
2. Heffler KF et al. JAMA Pediatrics. 2020;174(7):690-696. PMID 32310265 • status: allowed-list • what we read: secondary-only
3. Kushima M et al. JAMA Pediatrics. 2022;176(4):384-391. doi:10.1001/jamapediatrics.2021.5778 • status: allowed-list • what we read: secondary-only
4. Takahashi I et al. JAMA Pediatrics. 2023;177(10):1039-1046 • status: allowed-list • what we read: secondary-only
5. Brushe ME et al. JAMA Pediatrics. 2024;178(4):369-375 • status: allowed-list • what we read: secondary-only
6. Madigan S et al. JAMA Pediatrics. 2019;173(3):244-250 • status: allowed-list • what we read: secondary-only
7. World Health Organization. Guidelines on physical activity, sedentary behaviour and sleep for children under 5 years of age. 2019 • status: allowed-list • what we read: secondary-only
8. American Academy of Pediatrics. Media and Young Minds. Pediatrics. 2016;138(5) • status: allowed-list • what we read: secondary-only

9. Zamfir MT. Journal of Romanian Literary Studies. 2018;(13) · status: VERIFY · what we read: secondary-only
10. Marcelli D, Bossière MC, Ducanda AL. Enfances & Psy. 2018;79(3):142-160 · status: VERIFY · what we read: secondary-only
11. Heffler KF, Oestreicher LM. Medical Hypotheses. 2016;91:114-122 · status: VERIFY · what we read: secondary-only
12. Ophir Y et al. JAMA Network Open. 2023;6(12):e2346775. doi:10.1001/jamanetworkopen.2023.46775 · status: VERIFY · what we read: secondary-only
13. Lin P, Wu WT, Guo YL. JAMA Pediatrics. 2025;179(1):90-91. doi:10.1001/jamapediatrics.2024.4432 · status: VERIFY · what we read: secondary-only
14. Melchior M et al. BMC Public Health. 2022;22:865. doi:10.1186/s12889-022-13296-5 · status: VERIFY · what we read: secondary-only
15. Takahashi N, Tsuchiya KJ, et al. Psychiatry Research. 2023 · status: VERIFY · what we read: secondary-only
16. Cai C et al. Brain and Behavior. 2025. doi:10.1002/brb3.70884 · status: VERIFY · what we read: secondary-only
17. Zhang Y et al. JAMA Network Open. 2023 · status: VERIFY · what we read: secondary-only
18. Sundarimaa E et al. Journal of Autism and Developmental Disorders. 2025 · status: VERIFY · what we read: secondary-only
19. Hill MM, Gangi DN, Miller M. Child Psychiatry & Human Development. 2024. doi:10.1007/s10578-024-01785-0 · status: VERIFY · what we read: secondary-only
20. Yuan G et al. Journal of Autism and Developmental Disorders. doi:10.1007/s10803-024-06665-z · status: VERIFY · what we read: secondary-only
21. Liu H et al. Reviews on Environmental Health. 2025;40(2):437-444 · status: VERIFY · what we read: secondary-only
22. Chonchaiya W et al. Acta Paediatrica. 2011;100(7):1033-1037 · status: VERIFY · what we read: secondary-only
23. Patterns of early exposure to screens by children with ASD. Discover Mental Health. 2025;5(1):211 · status: VERIFY · what we read: secondary-only
24. Slobodin O, Heffler KF, Davidovitch M. J Dev Behav Pediatr. 2019;40(4) · status: VERIFY · what we read: secondary-only
25. Montes G. Academic Pediatrics. 2016;16(2):122-128. doi:10.1016/j.acap.2015.08.007 · status: VERIFY · what we read: secondary-only
26. Yamamoto M et al. JAMA Pediatrics. 2023;177(11):1168-1175. doi:10.1001/jamapediatrics.2023.3643 · status: VERIFY · what we read: secondary-only
27. Sadeghi S et al. Early Child Development and Care. 2021;191(6):827-836. doi:10.1080/03004430.2019.1649256 · status: VERIFY · what we read: secondary-only
28. Apims 2023; Heffler 2022; Hermawati 2018 (case reports and uncontrolled series) · status: VERIFY · what we read: secondary-only
29. Detroja S, Bhatia G. Indian J Psychol Med. 2024. doi:10.1177/02537176241263310 · status: VERIFY · what we read: secondary-only
30. Krijnen LJG et al. Autism. 2026. doi:10.1177/13623613261434478 · status: VERIFY · what we read: secondary-only
31. van Asselt A. 2026. doi:10.1177/25739581261452097 · status: VERIFY · what we read: secondary-only
32. Dhungel O, Pokhrel M, Karki U. JNMA. 2026;64(297) · status: VERIFY · what we read: secondary-only
33. Bundesverband autismus Deutschland e.V. 'Virtueller Autismus' position statement · status: VERIFY · what we read: secondary-only
34. Science Media Centre (UK). Expert reaction, April 2020 · status: VERIFY · what we read: secondary-only
35. Psihologia.ro; psychologiescientifique.org (early critiques) · status: VERIFY · what we read: secondary-only
36. Spitzer M. Nervenheilkunde. 2023;42:332-341. doi:10.1055/a-2022-0301 · status: VERIFY · what we read: secondary-only
37. Ozyazici K. Acta Psychologica. 2026;268 · status: VERIFY · what we read: secondary-only
38. Madigan S et al. JAMA Pediatrics. 2020;174(7):665-675; Mallawaarachchi S et al. JAMA Pediatrics. 2024;178(10):1017-1026 · status: VERIFY · what we read: secondary-only
39. American Academy of Pediatrics. Digital Ecosystems, Children, and Adolescents. Pediatrics. 2026;157(2):e2025075320 · status: VERIFY · what we read: secondary-only

Questions parents ask about "virtual autism" and screens

Draft status: draft: secondary sources only; not cleared to publish · **Last reviewed:** 2026-09-28 · **Words:** 2,574 (clinician scope 1,031) · **Source file:** content/research-hub/faq.md · **SHA-256:** 97cccd31e3c3

FAQ-001 "Virtual autism" is a term some clinicians use for autism-like behaviors seen in some young children with heavy early screen exposure. It is not a medical diagnosis. Studies have found links (associations) between early screen time and later outcomes, not proof that screens cause autism, and autism is not a parent's fault. If you're worried about your child, talk with your pediatrician and ask about a free early intervention evaluation.

FAQ-002 **This is not medical advice.** These answers explain research. They can't tell you anything about your own child. Please take any worry about your child's development to your pediatrician or another qualified professional.

About the term SENSITIVITY FOCUS

1. What is "virtual autism"?

FAQ-003 It is a label some clinicians and writers use for autism-like behaviors reported in some young children who had a lot of screen time early in life. It was coined by a Romanian psychologist in 2018 ([Zamfir 2018](#) [VERIFY]) and carried into the research literature by a French child psychiatrist's hypothesis paper in 2019 ([Harlé 2019](#)). It is an idea that is being debated. It is not an established condition.

2. Is "virtual autism" a real diagnosis?

FAQ-004 No. It is not in the World Health Organization's ICD or in the DSM, which are the manuals doctors use. Even the psychologist who coined it says it is not a medical diagnosis ([VERIFY]). A 2024 review concluded that the evidence is too preliminary for it to be recognized as a diagnosis ([Detroja and Bhatia 2024](#) [VERIFY]). A doctor can diagnose autism, or a language or developmental delay, but not "virtual autism".

3. Do screens cause autism?

FAQ-005 No study has shown that they do. Some studies find that more early screen time **goes with** somewhat higher autism-screening scores, or with more autism diagnoses in some groups (for example, in boys but not girls in a large Japanese study: [Kushima 2022](#)). But:

- FAQ-006 • the largest review we have found reported an overall positive link, but it shrank and was no longer statistically significant after correcting for publication bias ([Ophir 2023](#) [VERIFY]);
- FAQ-007 • children with a higher genetic likelihood of autism already watch more as toddlers, which suggests screen use may be an early sign rather than a cause ([Takahashi N 2023](#) [VERIFY]);
- FAQ-008 • studies designed to get closer to cause found no clear causal signal, though each has important limits ([Lin 2025](#) [VERIFY]; [Cai 2025](#) [VERIFY]).

FAQ-009 Researchers have not ruled out that screens play some part for some children. They also have not shown that they do.

4. Is one particular show or video channel the problem?

FAQ-010 The studies we have found measured how much time children spent with screens, and sometimes how screens were used (alone or together, or a TV left on in the background). They did not compare particular shows or channels [\[VERIFY across the library before publishing\]](#), so they can't tell you anything about one show. We don't comment on particular shows or channels. If you're worried about your child, the next step is the same whatever they watch: talk with your pediatrician and ask about a free evaluation. Asking is a caring step, and it's not your fault.

5. My child watched a lot of TV as a baby and is now autistic. Did I do this?

FAQ-011 No. Autism is not a parent's fault. Researchers and autism organizations have warned that the "virtual autism" idea risks reviving the old, false belief that autism comes from parenting ([autismus Deutschland \[VERIFY\]](#); [Krijnen 2026 \[VERIFY\]](#)). Some studies suggest that children later identified as autistic may be drawn to screens early, and parents of a child who is harder to settle may lean on them more. That is a normal response. No study has shown that screens cause autism. Your child is the same child you love, and what matters now is understanding how they experience the world and what helps them thrive.

FAQ-012 If someone suggests you caused your child's autism, you can simply say: "No study has shown that screens cause autism, and autism isn't caused by parenting."

6. Where does the idea come from?

FAQ-013 From case histories of Romanian children who were already diagnosed with autism ([Zamfir 2018 \[VERIFY\]](#)), from French clinicians' observations in 2017 and 2018 ([Marcelli 2018 \[VERIFY\]](#)), and from a 2019 hypothesis paper that pulled together case reports from several countries ([Harlé 2019](#)). None of these tested the idea with a comparison group. See [where the term came from](#).

7. Is it the same as "electronic screen syndrome", "EPEE" or "digital autism"?

FAQ-014 They are related labels, and none is a recognized diagnosis. "EPEE" is a French proposal that deliberately avoids the word autism ([Marcelli 2018 \[VERIFY\]](#)). "Electronic screen syndrome" comes from a popular book and is broader than autism-like signs ([Dunckley \[VERIFY\]](#)). "Post-digital nannyng autism syndrome" was proposed by one research group ([Pouretamad 2022 \[VERIFY\]](#)). See the [glossary](#).

About the research

8. Why do some studies find a link and others don't?

FAQ-015 Studies measure different things (a yes/no TV question vs. hours a day; a screening checklist vs. a diagnosis), at different ages, in different countries, with different ways of accounting for family circumstances. Small studies with striking results are also more likely to be published. When one review corrected for that, the overall link was no longer statistically significant ([Ophir 2023 \[VERIFY\]](#)). And a very large French study found screen use went with slightly higher odds of a *medium* screening score but *lower* odds of a *high-risk* one ([Melchior 2022 \[VERIFY\]](#)).

9. What is "reverse causation", and why does it matter here?

FAQ-016 It means the arrow may point the other way. Instead of screens leading to autism-like signs, early autistic traits may lead to more screen time: a baby who finds screens especially soothing or predictable, or a family who, understandably, uses screens more with a child who connects with people in different ways. Genetic research supports this possibility ([Takahashi N 2023 \[VERIFY\]](#)). See [why a link is not proof of cause](#).

10. What about the stories of children who "got better" when screens were taken away?

FAQ-017 Those reports are real experiences, and we understand why they stand out to families. But they can't tell us *why* a child changed. In every published report we found, less screen time came with more adult attention and often professional support, there was no comparison group, and young children change quickly as they grow (Heffler 2022 [VERIFY]; Pakistan 2023 [VERIFY]). We have not found a randomized trial that tests this (Ozyazici 2026 [VERIFY]).

FAQ-018 It also helps to know what these reports measured: scores on autism checklists or assessments. A lower score is not the same as a child being happier or better off, and support for an autistic child is about their communication, comfort and well-being, not about making them seem less autistic.

11. What does the research say about screens and talking?

FAQ-019 Separate from the autism question, studies find that more early screen time goes with less back-and-forth talk at home and with weaker language on screening questionnaires. In home recordings, screen time at 36 months went with less talk: scaled to the average amount of daily screen time in the study, 1,139 fewer adult words, 843 fewer child vocalizations and 194 fewer conversational turns a day (Brushe 2024). More screen time at age 1 went with higher odds of communication and problem-solving delays at 2 and 4 (Takahashi I 2023). These are associations, and they are not autism studies.

If you're worried CLINICIAN SCOPE

12. How would a doctor tell autism from difficulties related to heavy screen use?

FAQ-020 We can't answer that for any child, and neither can a website. Only a qualified clinician who assesses your child, often over time, can. What we can say is that researchers have not agreed on any reliable way to separate a "screen-related" picture from autism, and the term itself is not a diagnosis. If you have concerns, the path is the same either way: talk with your pediatrician and ask for an evaluation.

13. Should I cut screens first and wait to see if things change before asking for an evaluation?

FAQ-021 Please don't wait. You can change screen habits *and* ask for an evaluation at the same time. If you notice changes, tell the people evaluating your child: what you see at home is useful to them. In the US, an early intervention evaluation for children under 3 is free [VERIFY], and early support is worth having whatever the reason for a child's difficulties. [How to ask for an evaluation](#).

FAQ-022 If your child has stopped using words or skills they used to have, tell your pediatrician soon rather than waiting for the next routine visit [VERIFY wording against CDC and AAP guidance; clinician reviewer to confirm].

14. What is early intervention, and is it really free?

FAQ-023 In the US, every state runs an early intervention program for babies and toddlers under 3 who have, or may have, developmental delays, under a federal law called IDEA (Part C). The evaluation is free to families. Whether ongoing services cost anything depends on your state; some use sliding fees or insurance, and services can't be refused because a family can't pay. You can ask for the evaluation and meetings in your family's language, with an interpreter if you need one [VERIFY all against official program pages]. See [early intervention](#).

15. Do I need a doctor's referral for early intervention?

FAQ-024 In the US, generally no. Parents can usually contact their state or local early intervention program directly [VERIFY]. Grandparents and other caregivers can raise a concern too; a parent or legal guardian usually needs to give consent for the evaluation [VERIFY]. Telling your pediatrician is still a good idea, so everyone is working together.

16. What if my child is 3 or older?

FAQ-025 In the US, you can ask your local public school district to evaluate your child for free, even if your child isn't in school yet [\[VERIFY\]](#). Put the request in writing and keep a copy. See [early intervention](#).

17. We don't live in the US. Where do we start?

FAQ-026 Start with your family doctor, pediatrician, or child and family health nurse (in the UK, your health visitor). Ask what free early support is available locally and whether you can refer yourself. We have general pointers for the [UK](#), [Canada](#) and [Australia](#).

Everyday life with screens CLINICIAN SCOPE

18. How much screen time is OK for a baby or toddler?

FAQ-027 The World Health Organization recommends no screen time for babies under 1 and no more than 1 hour a day for 2-to-4-year-olds, with plenty of active play ([WHO 2019](#)). For 1-year-olds, it is recalled as not recommending sedentary screen time at all [\[VERIFY against the guideline\]](#). The American Academy of Pediatrics gave detailed advice for under-5s in 2016 ([AAP 2016](#)) and published a new policy statement in January 2026 that reportedly replaces its earlier media guidance ([AAP 2026](#) [\[VERIFY\]](#)). These limits are precautionary. They are not a line past which harm is proven.

19. Do video calls with grandparents count?

FAQ-028 Video chat is live and two-way, which is different from watching a show. The AAP's 2016 statement is recalled as making an exception for video chatting even for very young babies [\[VERIFY\]](#). We will update this answer once we have read the AAP's 2026 statement.

20. Does background TV matter?

FAQ-029 Research suggests it may. Reviews found that more background TV went with weaker language and poorer thinking and learning outcomes ([Madigan 2020](#) [\[VERIFY\]](#); [Mallawaarachchi 2024](#) [\[VERIFY\]](#)). These are associations, but turning the TV off when no one is watching is an easy change for many families.

21. Does watching together help?

FAQ-030 Watching *with* a caregiver who talks about what's on screen went with stronger language and better thinking outcomes in reviews of many studies ([Madigan 2020](#) [\[VERIFY\]](#); [Mallawaarachchi 2024](#) [\[VERIFY\]](#)). If screens are part of your day, sitting together and chatting about them is a good habit.

22. My autistic child loves their tablet. Is that harmful?

FAQ-031 That depends on your child, and it's a good question for the professionals who know them. Autistic researchers point out that screens can meet real needs, such as predictability, calm and connection, and that research should weigh benefits as well as risks ([van Asselt 2026](#) [\[VERIFY\]](#)). And if your child uses a tablet or device to communicate (AAC), that is a communication tool, not "screen time". Keep it within reach.

23. What can we do at home, whatever the research shows?

FAQ-032 Talk, sing, read and play together as part of ordinary family life. Not as a treatment, and not as a test. Pause and wait for your child's turn, say what you see, repeat what they say and add one word, offer choices (a point, a reach or a look counts as an answer), follow their lead, and sing songs with actions. Talk, sing and read in the language you know best. Every language counts. If you'd like ideas, our free printable, [Five 5-Minute Plays](#), has quick screen-free games for little ones, just for fun and family time.

For autistic readers SENSITIVITY FOCUS

24. I'm autistic, or my child is. How should I read this research?

FAQ-033 With care, and knowing where we stand. Autism is a lifelong difference, not damage and not something a family did. Much of this research uses clinical words such as "disorder", "symptoms", "severity" and "risk"; we quote them to describe what a study measured, not to describe autistic people ([glossary](#)). Some of the strongest critiques of the "virtual autism" idea come from autism researchers ([Krijnen 2026](#) [VERIFY]) and from an autistic researcher writing from their own experience ([van Asselt 2026](#) [VERIFY, including that the author identifies as autistic in the paper itself]). See [a note to autistic readers and their families](#). If something on our pages is hurtful or wrong, please tell us.

About this site

25. Why do some sources say [VERIFY]?

FAQ-034 It means we have not yet been able to read that source's original abstract or full text ourselves, so the details are provisional. We would rather show you that than pretend to certainty. Each study page has a "What we read" section. See our [editorial policy](#).

26. You sell play products. Why should I trust this?

FAQ-035 You're right to ask. Play Before Pixels is a trade name of AlphaPlay LLC, and we sell books and printables about play. That is a conflict of interest, so we have rules: our research pages never link to products, never mention prices, and never suggest that anything we sell helps with autism or any condition. This page has one link to a free play printable. If you choose to join our email list there, those emails do include our products, and you can unsubscribe at any time [final wording follows the hub email route the owner chooses]. We include critical and null research with the same care as supportive research. We are not clinicians. Read our [editorial policy](#) and judge for yourself.

27. I found a mistake. How do I tell you?

FAQ-036 Please use our contact form. You'll get an instant automatic reply, and a person reviews everything else within [5] business days. If we change a page because of a correction, we note the change and the date on that page. See [corrections](#).

FAQ-037 Worried? Talk with your pediatrician. Free early intervention evaluations are available for young children. Asking early is a caring step, and it's not your fault. [VERIFY access wording against official program pages at publish.] Outside the US: ask your doctor, nurse or health visitor about free local early support.

SOURCES THIS PAGE LISTS (FROM ITS FRONT MATTER)

1. See each linked study page for its full citation and what we read.
2. Harlé B. 2019; Heffler KF et al. 2020; Kushima M et al. 2022; Takahashi I et al. 2023; Brushe ME et al. 2024; WHO 2019; AAP 2016 (allowed-citation list) · [what we read: secondary-only](#)
3. Ophir 2023; Lin 2025; Takahashi N 2023; Cai 2025; Melchior 2022; Krijnen 2026; van Asselt 2026; autismus Deutschland; Madigan 2020; Mallawaarachchi 2024; AAP 2026; IDEA Part C program information [all VERIFY] · [what we read: secondary-only](#)

Glossary

Draft status: draft: not cleared to publish until reviewed with the pages it supports · **Last reviewed:** 2026-09-28 · **Words:** 2,272 (clinician scope 877) · **Source file:** content/research-hub/glossary.md · **SHA-256:** c3419fb5f2d9

GLO-001 Plain-language meanings of words you'll meet on our research pages. Where it helps, we point to a study that shows the idea in action.

GLO-002 "Virtual autism" is a term some clinicians use for autism-like behaviors seen in some young children with heavy early screen exposure; it is not a medical diagnosis. Studies show links, not proof of cause.

Words about the debate SENSITIVITY FOCUS

GLO-003 **"Virtual autism."** A term some clinicians use for autism-like behaviors reported in some young children with heavy early screen exposure. It is **not a medical diagnosis** and is not in the ICD or DSM. Coined in Romania in 2018 ([Zamfir 2018](#) [VERIFY]) and proposed in the research literature in 2019 ([Harlé 2019](#)). Also seen as "autisme virtuel" (French), "autismul virtual" (Romanian), "virtueller Autismus" (German), "sanal otizm" (Turkish) and "autismo virtual" (Spanish and Portuguese).

GLO-004 **"Digital autism."** Another informal label for the same idea. Not a diagnosis.

GLO-005 **EPEE (Exposition précoce et excessive aux écrans).** "Early and excessive screen exposure": a set of signs proposed by French clinicians in 2018. The name deliberately avoids the word autism. Not a recognized diagnosis ([Marcelli 2018](#) [VERIFY]).

GLO-006 **Electronic Screen Syndrome (ESS).** A label from a popular book describing a proposed problem with mood, attention and arousal linked to interactive screens. Broader than autism-like signs. Not a recognized diagnosis ([Dunckley](#) [VERIFY]).

GLO-007 **Post-Digital Nannyng Autism Syndrome (PDNAS).** A label proposed by one Iranian research group. Not a recognized diagnosis ([Pouretamad 2022](#) [VERIFY]).

GLO-008 **"Screen-related developmental delay."** A more neutral term suggested by autism researchers in place of "virtual autism" ([Krijnen 2026](#) [VERIFY]).

GLO-009 **"Screen-induced developmental deviation."** A research term proposed by clinicians in Nepal. Note that "screen-induced" assumes a cause that has not been shown ([Dhungel 2026](#) [VERIFY]).

GLO-010 **Parent blame.** The old, false idea that autism is produced by how parents raise a child. Labels that put the cause in a family's home life, such as "digital nannyng", can carry the same message even when they are about screens. Autism organizations and researchers warn that "virtual autism" risks reviving it ([autismus Deutschland](#) [VERIFY]).

Words about autism and development CLINICIAN SCOPE SENSITIVITY FOCUS

GLO-011 **Autism (autism spectrum disorder, ASD).** A lifelong neurodevelopmental difference that affects how a person communicates, relates to others and experiences the world, diagnosed by qualified clinicians using established criteria [VERIFY definition against WHO ICD-11 or DSM-5-TR wording]. "ASD" (autism spectrum disorder) is the clinical term used in diagnoses and in most studies, so you may see it on your child's reports; that is normal and expected. On our pages we mostly say plain "autism", which many autistic people prefer, and use "ASD" when quoting a study or a clinical document.

- GLO-012 **Autistic.** Many autistic people prefer identity-first language ("autistic child") because they see autism as part of who they are. Others prefer "child with autism". We mostly use identity-first language and respect each person's choice.
- GLO-013 **Autism-like behaviors / "autism-like symptoms".** Behaviors that resemble some features of autism, usually measured with a checklist. Having some of them does not mean a child is autistic.
- GLO-014 **Autistic traits.** Ways of communicating, playing, moving or sensing that are common in autistic people, such as intense interests, repetitive movements or a preference for sameness. Many studies call these "symptoms". We say "traits" or "signs", because they are part of how a person experiences the world, not signs of illness.
- GLO-015 **"Severity" scores.** Some checklists and assessments give a "severity" score: roughly, how many autistic traits were observed, and how strongly, on that tool. We use the word only in quotation marks. A higher score is not a measure of a child's well-being, happiness or worth, and a lower score after some change does not mean a child is better off.
- GLO-016 **"Risk" and "likelihood".** Studies and screening tools often talk about "risk of autism" or a "high-risk" screening score. In research, "risk" just means the chance of an outcome. Because autism is a difference and not a danger, we prefer "likelihood" (for example, "increased likelihood of autism"), a word some autism researchers also use (Hill 2020 [VERIFY]). When a study or tool uses "risk", we quote it.
- GLO-017 **Neurodiversity / neurodiversity-affirming.** The idea that brains naturally differ, and an approach that respects autistic and other neurodivergent ways of being rather than trying to make people "normal" (van Asselt 2026 [VERIFY]).
- GLO-018 **Developmental delay.** When a child reaches skills (talking, moving, playing, problem-solving) later than most children the same age. It has many possible reasons and is not the same as autism.
- GLO-019 **Loss of skills (regression).** When a child stops using words or skills they used to have. At any age, it is a reason to tell your pediatrician soon rather than waiting for the next routine visit [VERIFY wording against CDC and AAP guidance].
- GLO-020 **AAC (augmentative and alternative communication).** Ways to communicate besides speech, including picture boards and speech-generating apps on tablets. An AAC device is a communication tool, not "screen time".
- GLO-021 **Conversational turns.** Back-and-forth exchanges between a child and an adult: the child babbles, the adult answers, the child answers again. Counted in home recordings (Brushe 2024).
- GLO-022 **Co-viewing.** Watching a screen together with a caregiver who talks about it. Associated with stronger language in reviews (Madigan 2020 [VERIFY]).
- GLO-023 **Background TV.** A TV on in the room when no one is really watching. Associated with weaker language in reviews (Madigan 2020 [VERIFY]).

Tests and checklists CLINICIAN SCOPE

- GLO-024 **Screening vs. diagnosis.** A **screening** tool is a short questionnaire that flags children who may need a closer look. A **diagnosis** comes from a full assessment by a qualified clinician. Most studies on our hub use screening tools, not diagnoses.
- GLO-025 **M-CHAT-R/F (Modified Checklist for Autism in Toddlers, Revised with Follow-Up).** A parent questionnaire used to screen toddlers for signs that suggest a fuller autism assessment [VERIFY]. Its results are grouped as low, medium or high "risk" (see "Risk" and "likelihood" above). A positive result is not a diagnosis (Sundarimaa 2025 [VERIFY]).

- GLO-026 **ADOS (Autism Diagnostic Observation Schedule).** A structured, play-based assessment used by trained clinicians as part of an autism evaluation [\[VERIFY\]](#).
- GLO-027 **CARS (Childhood Autism Rating Scale).** A rating scale clinicians use to describe autism-related behaviors [\[VERIFY\]](#).
- GLO-028 **SCQ (Social Communication Questionnaire).** A parent questionnaire about social communication, used for screening [\[VERIFY\]](#).
- GLO-029 **ASQ-3 (Ages and Stages Questionnaires, Third Edition).** A parent-completed developmental screening questionnaire covering areas like communication, problem-solving and motor skills [\[VERIFY\]](#). Not an autism test ([Takahashi I 2023](#)).

How studies are designed

- GLO-030 **Case report / case series.** A description of one child or a small group. Useful for raising questions; cannot show cause ([Heffler 2022](#) [\[VERIFY\]](#)).
- GLO-031 **Cross-sectional study.** Measures everything at one point in time. Can show that two things go together, not which came first ([Sundarimaa 2025](#) [\[VERIFY\]](#)).
- GLO-032 **Case-control study.** Compares children who have a condition (cases) with children who don't (controls), often asking about the past. Vulnerable to memory bias ([Chonchaiya 2011](#) [\[VERIFY\]](#)).
- GLO-033 **Cohort study.** Follows a group over time. Better than a one-time survey for knowing what came first, but still observational ([Kushima 2022](#)).
- GLO-034 **Observational study.** Researchers measure what people already do, without assigning anything. Most research on screens and autism is observational.
- GLO-035 **Randomized controlled trial (RCT).** Participants are assigned by chance to one approach or another, which balances other factors between groups. The strongest way to test cause. We have not found one that tests screen reduction and autism-like signs in young children ([Ozyazici 2026](#) [\[VERIFY\]](#)).
- GLO-036 **Blinding.** When the people measuring the outcome don't know which group a child is in. Without it, expectations can colour the results.
- GLO-037 **Systematic review.** A review that searches for all relevant studies using stated rules, so it is less likely to cherry-pick ([Slobodin 2019](#) [\[VERIFY\]](#)).
- GLO-038 **Meta-analysis.** A systematic review that combines the numbers from many studies into one estimate ([Ophir 2023](#) [\[VERIFY\]](#)).
- GLO-039 **Narrative review.** An expert's overview of a topic without a systematic search. Useful, but more open to the author's views.
- GLO-040 **Hypothesis paper.** A paper that proposes an idea to be tested, rather than testing it ([Harlé 2019](#)).
- GLO-041 **Grey literature.** Material not published in peer-reviewed journals: web posts, handouts, dissertations, reports.
- GLO-042 **Peer review.** Checking of a paper by independent experts before a journal publishes it. It reduces errors but doesn't guarantee a study is right.
- GLO-043 **Preprint.** A paper shared before peer review ([example](#)).
- GLO-044 **Letter / commentary.** A short published response or opinion, usually without new data ([Alper 2020](#) [\[VERIFY\]](#)).

Words about cause and chance

- GLO-045 **Association (link, correlation).** Two things tend to occur together. It does not tell you whether one produces the other.
- GLO-046 **Causation.** One thing actually produces a change in another. Showing it usually needs trials or very careful designs that rule out other explanations.
- GLO-047 **Reverse causation.** The arrow points the other way from what you'd assume. Here: early autistic traits leading to more screen time, rather than screen time leading to autistic traits ([Takahashi N 2023](#) [VERIFY]).
- GLO-048 **Confounding / confounder.** A third factor linked to both things being studied that makes them look connected. Family stress, income, parents' own traits and genes can all be confounders.
- GLO-049 **Genetic confounding.** When shared genes influence both the "exposure" (for example screen habits) and the outcome, making them look linked ([Zhang 2023](#) [VERIFY]).
- GLO-050 **Recall bias.** Errors that come from asking people to remember the past, especially after something has happened (such as a diagnosis) that may shape what they remember.
- GLO-051 **Selection bias.** When the people in a study differ from the wider population in ways that affect the results, for example families recruited online or at a specialist clinic.
- GLO-052 **Ecological fallacy.** Assuming that a pattern across places (counties, countries) applies to individuals ([Waldman 2008](#) [VERIFY]).
- GLO-053 **Dose-response.** When more of an exposure goes with steadily more of an outcome. A dose-response pattern makes a link more convincing, but it still doesn't prove cause. In one large study the odds rose across the lower screen-time groups and then levelled off, with similar, overlapping estimates in the two highest groups ([Kushima 2022](#)).
- GLO-054 **Odds ratio (OR).** A way of comparing how likely an outcome is in two groups. An OR of 1 means no difference; above 1 means more likely; below 1, less likely. Some studies report the **log odds ratio**, where 0 means no difference.
- GLO-055 **Confidence interval (CI).** The range of values the true result plausibly falls in. A wide interval means the estimate is uncertain. An interval from 1.55 to 660.8 means the study had too little data to say much ([Rangaraj 2026](#) [VERIFY]).
- GLO-056 **Statistically significant.** A result that would be unlikely if there were truly no effect, judged by a conventional threshold (usually $P < .05$). It says nothing about how big or important an effect is, and a non-significant result is not proof of no effect.
- GLO-057 **Publication bias.** Studies with striking or positive results are more likely to be published, which can make a link look stronger than it is.
- GLO-058 **Trim-and-fill.** A statistical way of estimating what a meta-analysis would show if missing (unpublished) studies were included. It rests on assumptions ([Ophir 2023](#) [VERIFY]).
- GLO-059 **Instrumental variable (IV) analysis.** A method that uses a third factor, assumed to affect the outcome only through the exposure, to get closer to cause. Its results are only as good as that assumption ([Lin 2025](#) [VERIFY]).
- GLO-060 **Mendelian randomization (MR).** An instrumental-variable method that uses genetic variants as a natural experiment ([Cai 2025](#) [VERIFY]).
- GLO-061 **Polygenic risk score (PRS).** A number that sums many small genetic differences linked to a trait or condition. It explains only a small part of any one person's likelihood ([Takahashi N 2023](#) [VERIFY]).

- GLO-062 **Pleiotropy.** When the same genes affect several traits, which can confuse genetic studies.
- GLO-063 **Mediation analysis.** A statistical estimate of how much of a link "passes through" a third factor. In cross-sectional data it describes numbers, not proven pathways ([Chen 2020](#) [\[VERIFY\]](#)).
- GLO-064 **Maturation.** Children's ordinary development over time. It can make any change after a new approach look like an effect of that approach, unless there is a comparison group.
- GLO-065 **Regression to the mean.** Things measured at an unusually high or low point tend to move back toward average next time, with or without any change in circumstances.

Help and services [CLINICIAN SCOPE](#)

- GLO-066 **Early intervention (EI).** In the US, services for babies and toddlers under 3 with, or at risk of, developmental delays, run by each state under IDEA Part C. The evaluation is free [\[VERIFY\]](#). See [early intervention](#).
- GLO-067 **IDEA.** The US Individuals with Disabilities Education Act. **Part C** covers birth to 3; **Part B** covers ages 3 to 21, including preschool special education through public school districts [\[VERIFY\]](#).
- GLO-068 **IFSP (Individualized Family Service Plan).** The written plan made with a family if a child under 3 qualifies for early intervention in the US [\[VERIFY\]](#).
- GLO-069 **IEP (Individualized Education Program).** The written plan for a child aged 3 or older who qualifies for special education services in the US [\[VERIFY\]](#).
- GLO-070 **Health visitor.** In the UK, a specialist nurse who supports families with young children and does routine development reviews [\[VERIFY\]](#).
- GLO-071 **Speech-language pathologist (SLP).** A professional who assesses and supports communication: talking, understanding, social communication and AAC. Called a speech and language therapist in the UK and a speech pathologist in Australia [\[VERIFY titles\]](#).

About our sources

- GLO-072 [\[VERIFY\]](#). We have not yet read the original abstract or full text of this source ourselves, so the details are provisional. See our [editorial policy](#).
- GLO-073 **What we read.** Each study page says whether we read the full text, the abstract, or only secondary summaries of the source.

GLO-074 **This is not medical advice.** These are general explanations of research words. For questions about your own child, talk with your pediatrician or another qualified professional.

- GLO-075 Worried? Talk with your pediatrician. Free early intervention evaluations are available for young children. Asking early is a caring step, and it's not your fault. [\[VERIFY access wording against official program pages at publish.\]](#) Outside the US: ask your doctor, nurse or health visitor about free local early support.

SOURCES THIS PAGE LISTS (FROM ITS FRONT MATTER)

1. Definitions are general research vocabulary written in plain language by the editors. Examples point to the study pages where each term is used.
2. Instrument and program names (M-CHAT-R/F, ADOS, CARS, SCQ, ASQ-3, IDEA Part C/Part B) [\[VERIFY each definition against the instrument publisher or official program page before publishing\]](#) · what we read: secondary-only

Questions to ask your pediatrician

Draft status: draft: needs clinician review and autistic sensitivity read before publishing · **Last reviewed:** 2026-09-28 · **Words:** 593 (clinician scope 593) · **Source file:** content/research-hub/pediatrician-questions.md · **SHA-256:** 99cf8d77fc27

PED-001 | *Print this, circle the questions that matter to you, and bring it to your child's next visit. You don't have to ask them all. Start the visit with the worry that matters most to you.*

PED-002 | **This is not medical advice.** It's a list of questions to help you talk with your child's doctor. Only a professional who meets your child can answer them for your family.

PED-003 | **Child's age:** _____ months **Date of visit:** _____

PED-004 | **Language(s) we speak at home:** _____ (If you'd like an interpreter, ask the clinic whether one is available.)

Before the visit: my notes

PED-005 | What worries me (what I see, when, how often):

PED-006 | _____

PED-007 | _____

PED-008 | Anything my child used to do but has stopped doing: _____

PED-009 | Examples or short videos I can show the doctor: _____

PED-010 | What my child is good at and enjoys: _____

About my child's development

PED-011 | ☐ Is my child's **talking and understanding** where you'd expect for their age?

PED-012 | ☐ How does my child's **play, pointing, eye contact and response to their name** look to you?

PED-013 | ☐ What skills would you expect to see **in the next few months**?

PED-014 | ☐ Is there anything you'd like me to **watch for** between now and the next visit?

PED-015 | ☐ Has my child's **hearing** been checked? Should it be checked again?

PED-016 | ☐ Should my child's **vision** be checked?

About screening

PED-017 | ☐ Has my child had a **developmental screening** or an **autism screening** questionnaire? What did it show?

PED-018 | ☐ What does a positive (or clear) screening result **mean**, and what doesn't it mean?

PED-019 | ☐ If the screen is clear but I'm still worried, **what's the next step**?

About evaluation and support

PED-020 | ☐ Would a **full evaluation**, or a **speech and language evaluation**, be helpful? Who does it, and how long is the wait?

PED-021 | ☐ (US, under 3) Can you help me contact our **early intervention** program, or can I refer my child myself?

PED-022 | ☐ (US, 3 and older) Should I ask our **public school district** for an evaluation?

- PED-023 ☐ (Outside the US) What **free local early support** is there, and can I refer myself?
- PED-024 ☐ If support is offered, **what would it look like day to day**, and how does it fit my child's comfort and the way they communicate?
- PED-025 ☐ While we wait, **what can we do** at home?
- PED-026 ☐ Is there **anything I should not wait** on?

About screens and daily life

- PED-027 ☐ How do our family's **screen habits** fit with my child's development? Is there anything you'd suggest we change?
- PED-028 ☐ I've read about "**virtual autism**". What do you think about the research? *(It is a term some clinicians use; it is not a medical diagnosis.)*
- PED-029 ☐ My child uses a **tablet or app to communicate**. How should we think about that? *(Communication devices are not "screen time".)*
- PED-030 ☐ How are my child's **sleep, eating and activity**, and do they affect any of this?

About us as a family

SENSITIVITY FOCUS

- PED-031 ☐ Where can we find **support and information for our family** while we learn more?
- PED-032 ☐ Can a **grandparent or another caregiver** who looks after my child come to visits or talk with you?
- PED-033 ☐ If my child is autistic, where can we **learn from autistic adults** and autistic-led groups?
- PED-034 ☐ Who should I **call** if things change or I have new worries before the next visit?

After the visit

- PED-035 Next steps: _____
- PED-036 Referrals made (name, phone, date): _____
- PED-037 Follow up by (date): _____ If I haven't heard by then, I will: _____

PED-038 *You know your child best. Asking questions is a caring step, and it's not your fault if you have worries.*

PED-039 **Outside the US:** ask your doctor, nurse or health visitor about free local early support.

PED-040 More plain-language research explanations: playbeforepixels.com/research/

PED-041 © 2026 AlphaPlay LLC. Play Before Pixels is a trade name of AlphaPlay LLC.

SOURCES THIS PAGE LISTS (FROM ITS FRONT MATTER)

1. Questions written by the editors to help families prepare for a visit. They make no factual claims. Background on screening and early intervention: see </research/early-intervention/> [program facts VERIFY].

How to ask for a free early intervention evaluation

Draft status: draft: every program fact is [VERIFY]; official program pages could not be opened on 2026-09-28 · **Last reviewed:** 2026-09-28 · **Words:** 1,450 (clinician scope 1,450) · **Source file:** content/research-hub/early-intervention.md · **SHA-256:** d37a1a1bccd5

EIV-001 **This is not medical or legal advice.** Programs and rules differ by state and country and change over time. Always check with your local program. Everything specific on this page is marked [VERIFY] until we have checked it against the official source.

EIV-002 If you're worried about how your child is talking, playing, moving, responding or connecting with you, **you can ask for help now.** You don't need to be sure anything is wrong, you don't need a diagnosis, and you don't need to wait and see. Asking early is a caring step, and it's not your fault. Grandparents and other caregivers can start this too [VERIFY].

EIV-003 This applies whether or not screens are part of the story. Support is useful whatever the reasons behind what you're noticing, and an evaluation is how you find out what your child needs.

In the United States: children under 3

What it is

EIV-004 Every US state runs an **early intervention** program for babies and toddlers from birth to age 3 who have, or may have, a developmental delay or disability. It exists under a federal law, the Individuals with Disabilities Education Act, **Part C** [VERIFY].

The key facts [VERIFY each against official pages]

- EIV-005 • **You can refer your own child.** In general, parents can contact the program directly. A doctor's referral is not required, though it's a good idea to tell your pediatrician.
- EIV-006 • **The evaluation is free.** Evaluation and assessment are provided at no cost to families.
- EIV-007 • **There are timelines.** Federal rules set a timeline (commonly described as 45 days from referral) for the evaluation, assessment and first planning meeting.
- EIV-008 • **If your child qualifies,** the family and the team write an **Individualized Family Service Plan (IFSP).** Services are often provided at home or in other everyday settings.
- EIV-009 • **Cost of services depends on your state.** Some services are free; some states use sliding fees or insurance. Federal rules say services can't be refused because a family can't pay.
- EIV-010 • **A diagnosis isn't needed** to be evaluated. Eligibility is based on your child's development, and each state sets its own definition of delay.
- EIV-011 • **Grandparents and other caregivers can raise a concern.** Anyone who looks after a child can contact the program. A parent or legal guardian, or someone acting as the child's parent, usually needs to give consent for the evaluation.
- EIV-012 • **You can use your own language.** You can ask for the evaluation and meetings in your family's language, with an interpreter if you need one.
- EIV-013 • **You stay in charge.** You are part of the team. You can ask what any suggested service involves and why, and you can accept some services and decline others.

How to ask, step by step

- EIV-014 1. **Find your state's program.** Use a national directory, such as the CDC's "Learn the Signs. Act Early." state contacts, or the list of state early intervention coordinators kept by the federally funded Early Childhood Technical Assistance Center [\[VERIFY links\]](#). Your pediatrician's office can also tell you.
- EIV-015 2. **Call, email or use the online form.** Say something like the script below.
- EIV-016 3. **Write down** the date, the person you spoke to, and what they said will happen next.
- EIV-017 4. **Follow up in writing** if you don't hear back within the time they gave you. Keep copies.
- EIV-018 5. **Tell your pediatrician** you've made the referral, so they can share any records the program needs.

A script you can use

EIV-019 "Hello. I'd like to refer my child for a free early intervention evaluation. My child is [age in months]. I'm concerned about [how they talk / how they play / how they respond when I call their name / how they move / something else]. I'd like to know the next steps and when I can expect to hear from you. My contact details are [phone/email]."

EIV-020 You don't need special words. "I'm worried and I'd like an evaluation" is enough.

What to bring or have ready

- EIV-021 • Your child's birth date, and any health or hearing and vision check results you have.
- EIV-022 • Notes or short examples of what worries you: what your child does, when, and how often, and anything your child used to do but has stopped doing.
- EIV-023 • If you like, a milestone checklist for your child's age. The CDC's "Learn the Signs. Act Early." program offers free ones, including in Spanish [\[VERIFY\]](#).
- EIV-024 • Any questions (our [printable list of questions for your pediatrician](#) may help).
- EIV-025 • If you want it, someone to come with you or join the call.

Children aged 3 and older

- EIV-026 In the US, children aged 3 and older can be evaluated through the **local public school district**, under **IDEA Part B** (preschool special education), even if they don't attend a school yet [\[VERIFY\]](#).
- EIV-027 • Contact your local public school district's special education or preschool office and **ask in writing** for a free evaluation for special education services. Date the request and keep a copy.
- EIV-028 • If your child is already in early intervention, the program should help plan the move to school-district services before your child turns 3 [\[VERIFY\]](#).
- EIV-029 • If your child qualifies, the plan is called an **Individualized Education Program (IEP)** [\[VERIFY\]](#).
- EIV-030 • Every state has at least one federally funded **Parent Training and Information Center** that helps families with these steps for free. The Center for Parent Information and Resources keeps a directory [\[VERIFY\]](#).

Talking with your pediatrician

- EIV-031 Early intervention and your pediatrician work best together. Your pediatrician can:
- EIV-032 • listen to your concerns and look at your child's development,

- EIV-033 • use developmental and autism screening questionnaires at routine visits (US pediatric guidance recommends developmental screening at set ages and autism screening at 18 and 24 months **[VERIFY against current AAP guidance]**),
- EIV-034 • check hearing and vision,
- EIV-035 • refer you to specialists for a full assessment if needed.
- EIV-036 • **A screening result is not a diagnosis.** A positive screen means "let's look more closely". A clear screen with ongoing worries is still worth raising again.

Outside the US

- EIV-037 • We can only give general pointers here. Please check your local services. **[VERIFY all]**

United Kingdom

- EIV-038 • Start with your **health visitor** or **GP**. In England, young children are offered routine health and development reviews, including one at around 2 to 2½ years, as part of the Healthy Child Programme **[VERIFY]**.
- EIV-039 • They can refer your child to local child development, speech and language, or community paediatric services. In many areas, parents can also **self-refer** to speech and language services **[VERIFY by area]**.
- EIV-040 • In England, each local authority publishes a **"Local Offer"** describing support for children with special educational needs and disabilities **[VERIFY]**. Scotland, Wales and Northern Ireland have their own systems **[VERIFY]**.

Canada

- EIV-041 • Start with your **family doctor, pediatrician or public health nurse**.
- EIV-042 • Services are run by each **province or territory**, and many offer free preschool speech and language or early-years developmental services that accept **parent self-referral** **[VERIFY by province/territory]**.
- EIV-043 • Your provincial or territorial health or children's services website will list what is available **[VERIFY]**.

Australia

- EIV-044 • Start with your **GP** or your **maternal and child health / child and family health nurse** **[VERIFY state names]**.
- EIV-045 • The **NDIS early childhood approach** supports young children with developmental concerns, and a diagnosis is not needed to ask for help. Check the current age range and how to contact an early childhood partner on the NDIS website **[VERIFY]**.

Everywhere else

- EIV-046 • Ask your doctor, nurse or health visitor about free local early support, and whether you can refer your child yourself.

Common worries **SENSITIVITY FOCUS**

- EIV-047 • **"What if I'm overreacting?"** Evaluators would much rather see a child who turns out not to need extra support than miss a child who would benefit from it. Asking is never wrong.
- EIV-048 • **"What if they say it's autism?"** Then you'll know more about how your child experiences the world, and you can get support that fits them. Autism is not a parent's fault, and your child is the same child they were before the evaluation. Autistic children grow, learn and thrive with understanding and the right help. Good support respects how your child communicates and what they love; it is not about making them seem less

autistic. Autistic adults' own writing and autistic-led organizations are good places to learn what autism is like from the inside.

EIV-049 **"We've had a lot of screen time. Will they judge us?"** Nearly every family uses screens. Evaluators are there to understand your child, not to judge you. It's fine to mention screen habits honestly; it helps the team see the whole picture. If you've noticed changes when your screen habits changed, say that too. What you've seen at home is useful information.

EIV-050 Worried? Talk with your pediatrician. Free early intervention evaluations are available for young children. Asking early is a caring step, and it's not your fault. **[VERIFY access wording against official program pages at publish.]** Outside the US: ask your doctor, nurse or health visitor about free local early support.

SOURCES THIS PAGE LISTS (FROM ITS FRONT MATTER)

1. Individuals with Disabilities Education Act (IDEA), Part C (infants and toddlers, birth to 3) and Part B section 619 (preschool, ages 3-5); implementing regulations 34 CFR Part 303 and Part 300 **[VERIFY against sites.ed.gov/idea]** · what we read: not read this session (site blocked); from editors' background knowledge
2. CDC 'Learn the Signs. Act Early.' program pages, including state early intervention contacts **[VERIFY at cdc.gov]** · what we read: not read this session (site blocked)
3. Early Childhood Technical Assistance Center (ECTA), list of state Part C coordinators **[VERIFY at ectacenter.org]** · what we read: not read this session (site blocked)
4. Center for Parent Information and Resources, directory of Parent Training and Information Centers **[VERIFY at parentcenterhub.org]** · what we read: not read this session
5. UK: NHS pages on the Healthy Child Programme reviews and health visitors; England SEND Local Offer; devolved-nation equivalents **[VERIFY]** · what we read: not read this session
6. Canada: provincial and territorial health and early-years ministries; Canadian Paediatric Society parent pages **[VERIFY]** · what we read: not read this session
7. Australia: NDIS early childhood approach; state maternal and child health services **[VERIFY at ndis.gov.au]** · what we read: not read this session

Editorial policy

Draft status: draft: needs legal review before publishing · **Last reviewed:** 2026-09-28 · **Words:** 1,259 · **Source file:** content/research-hub/editorial-policy.md · **SHA-256:** f9d5fb6a7e2e

POL-001 Our research hub explains what studies say about screens, early development and the term "virtual autism". Families make real decisions with information like this, so we hold ourselves to the standards below. If we fall short, please [tell us](#).

Who we are

POL-002 The research hub is published by **Play Before Pixels**, a trade name of **AlphaPlay LLC**. We make play materials for families: books and printables about talking, playing and reading together.

POL-003 • **We are not clinicians.** No one writing these pages is a doctor, psychologist or speech-language pathologist, and we never claim to be. We explain research; we do not assess children or give medical advice.

POL-004 • **We write as a team, not as a personality.** Research pages carry no personal stories and no founder story.

POL-005 • **AI assistance.** We use AI tools to help find, organize and draft material. Every page is checked by a person against the sources before it is published, and each study page says exactly what was read.

[VERIFY that this promise matches the publishing workflow before this page goes live.]

Our conflict of interest, stated plainly

POL-006 We sell play products. A site that sells things has a reason to make screens sound scary and play sound like a solution. We know that, so:

POL-007 • **Research pages never link to products**, never show prices, and never show product images or mascots.

POL-008 • Research pages may include **one** link to our free play printable, described as play and family time. **Never** as help for autism or any condition.

POL-009 • **Nothing we sell treats, prevents or relates to autism** or any developmental condition, and we never suggest otherwise.

POL-010 • We **never target** people searching for autism with ads, listings, hashtags or emails, in any language.

POL-011 • We accept **no money, sponsorship or free products** from device makers, app makers, EdTech companies or research authors for anything on the research hub. [VERIFY: confirm with the owner that this is and will remain true.]

POL-012 • We carry **no ads and no affiliate links** on research pages.

How we choose sources

POL-013 1. **Primary sources first.** We describe a study from the study itself: the abstract at minimum, and the full text wherever we can get it. Press releases and news articles are not enough.

POL-014 2. **We say what we read.** Every study page has a "What we read" section: *full text*, *abstract*, or *secondary only*. Secondary-only pages stay unpublished, or clearly marked [VERIFY], until the original is read.

POL-015 3. **An allowed-citation list.** A short list of sources has been checked and approved for use across our materials. Anything outside it is marked [VERIFY] until the original has been read and logged in our research log. [OWNER DECISION: see the note on review step 1.]

- POL-016 4. **Balance is a rule, not a courtesy.** We look for, and give equal space to, **critical, skeptical and null** studies and commentary, not only studies that report a link. In our [research library](#) the critical and null work is listed first.
- POL-017 5. **We weigh design, not headcount.** A large cohort study or a careful meta-analysis counts for more than many case reports. Each study page says how much weight we think it deserves, and why.
- POL-018 6. **Grey literature is labelled.** Web posts, handouts, preprints and position statements appear only when they matter to the history or the debate, and they are labelled as such.

How we write about findings

- POL-019 • **Associations are never described as causes.** We say "linked with" or "associated with". When a study's own title or conclusion claims a cause, we quote it, attribute it, and explain whether the design can support it.
- POL-020 • **Every study page covers limitations and "what it does NOT show".**
- POL-021 • **"Virtual autism" is always framed** as a term some clinicians use; not a medical diagnosis; associations, not causation.
- POL-022 • **Screening is not diagnosis.** We say which one a study measured.
- POL-023 • **Studies about talk or general development are never presented as autism studies.**
- POL-024 • **Numbers are quoted exactly as reported,** with confidence intervals where available, and without rounding that changes their meaning.
- POL-025 • **No fear, no blame.** We avoid alarm words and never suggest a parent caused their child's autism.

Respectful language SENSITIVITY FOCUS

- POL-026 • We describe autistic people with respect. We mostly use identity-first language ("autistic child"), which many autistic people prefer, and we honour individuals' and communities' own preferences. In translation we follow each language community's norms, checked with autistic reviewers.
- POL-027 • We never use words like "epidemic", "fight", "battle", "suffer", "cure" or "recover from autism".
- POL-028 • We never describe autism as damage, as an outcome to be prevented, reduced or reversed, or an autistic child as a worse outcome. When a study uses words such as "symptoms", "severity", "risk", "impairment" or "reversible", we put them in quotation marks, attribute them, and explain what was actually measured.
- POL-029 • A lower score on an autism checklist is never described as a child "getting better". We say what score changed.
- POL-030 • We don't describe parents' screen use as the cause of a child's autism, and we point out when a label (such as "digital nannying") implies it.
- POL-031 • Communication devices (AAC) are communication tools, not "screen time".
- POL-032 • We look for, and include with equal care, research and commentary by autistic authors and statements from autistic-led organizations, not only clinical studies about autistic people.
- POL-033 • Parents of autistic children are welcome here, and so are autistic readers, including autistic parents. Our main guide includes [a note to autistic readers and their families](#).

Review before publishing

- POL-034 Every research page goes through, in order:

- POL-035 1. **Source check.** A person reads each cited source (abstract at minimum) and confirms every number, quote and description. The page's "What we read" field is updated. [OWNER DECISION, content/research-hub/review/REVIEW-PACK.md section 9 item 1: this sentence and the "AI assistance" line above are true only if a human fact-checker is hired. Otherwise the claims and legal reviewer rewrites both to describe the AI-assisted check honestly. This page cannot publish with either sentence as it stands until that is decided.]
- POL-036 2. **Balance check.** Does the page give critical and null evidence the same care as supportive evidence?
- POL-037 3. **Autistic sensitivity read.** A paid autistic reviewer reads each new format before first publication.
- POL-038 4. **Medical-claims and legal review.** For the main guide, the early intervention page and anything giving practical advice. A clinician reviews the "what parents can do" sections.
- POL-039 5. **Compliance gate.** Our internal checklist (no health claims, no product links, no autism targeting, correct framing).
- POL-040 Each page shows a "**Last reviewed**" date.

Keeping pages up to date

- POL-041 • We re-review the main guide and the library at least **every six months**, and sooner when an important study is published. [VERIFY cadence with the owner.]
- POL-042 • New studies are added to the library whatever they find.
- POL-043 • When a study is corrected or retracted, we update its page and say so.

Corrections

- POL-044 If you think something is wrong, missing or unfair, please use our **contact form** and include the page and, if you can, the source.
- POL-045 • You'll get an **instant automatic reply**; a person reviews everything else within **[5] business days**.
- POL-046 • If we change a page because of a correction, we add a short **correction note** at the bottom with the date and what changed. We don't quietly rewrite.
- POL-047 • If we disagree, we'll tell you why.
- POL-048 We never ask for, and you should never send, information about your child's health.

Privacy

- POL-049 Our research pages do not collect information about children. The free printable linked from our research pages never asks for a child's name, age or birth date. [Final wording follows the owner's choice of hub sign-up route, ops/APPROVALS.md: a direct download, or an email only.] See our privacy policy.

-
- POL-050 © 2026 AlphaPlay LLC. Play Before Pixels is a trade name of AlphaPlay LLC.
-

SOURCES THIS PAGE LISTS (FROM ITS FRONT MATTER)

1. Play Before Pixels brand rules (brand/BRAND.md: hard rules 1-3 and 5; 'Autism searches'), ops/COMPLIANCE-GATE.md, marketing/CAMPAIGN-BIBLE.md section 4 (internal documents)

Comment table

Use one row per comment. Put the block ID from the left margin in the first column. Levels: must-fix, should-fix, suggestion. Add rows as needed, or use the same columns in a spreadsheet.

#	Block ID	Level	What the problem is	Suggested wording	Why
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					

Sign-off form A: autistic sensitivity read

Bundle version ID: _____ **Pages reviewed:** IDX, FAQ, GLO, PED, EIV, POL **Reviewer code:** SR-_____

1. I read the pages listed above in full, in the bundle version shown.
2. My comments are in the attached comment table: _____ must-fix, _____ should-fix, _____ suggestions.
3. Result (tick one):
 - ☐ **Passes** as written.
 - ☐ **Passes after changes:** it will pass once every must-fix item is resolved to my satisfaction in the re-check.
 - ☐ **Does not pass.** Main reason: _____
4. I reviewed wording, framing and respect only. I did not check medical, legal or research accuracy. I was not asked to review, and have not reviewed, any product.
5. My fee does not depend on my result. My answers to the independence questions are complete and true today.
Conflicts to note: _____
6. Credit (tick one):
 - ☐ No credit.
 - ☐ Unnamed credit: "Reviewed for respectful language by a paid autistic sensitivity reader on [date]."
 - ☐ Named credit, exactly as written here: _____
7. Only if I choose (tick any):
 - ☐ You may say in the named credit that I am autistic.
 - ☐ You may add my name to the page's structured data (reviewedBy).
8. I understand that my credit appears only on the pages I reviewed, only for this version, never with any product or in advertising, and that I can withdraw it at any time in writing.

Signature: _____ Date: _____

Re-check (within 30 days): version ID _____

- ☐ All my must-fix items are resolved.
- ☐ Not resolved: _____

Signature: _____ Date: _____

Sign-off form B: clinician review

Bundle version ID: _____ **Reviewer code:** CL-_____ **Credential:** _____ **License state and number (kept off the website):** _____

Scope reviewed: IDX opening boxes, "What we still don't know" and "What parents can do"; FAQ opening boxes and Q12–Q23; GLO "Words about autism and development", "Tests and checklists" and "Help and services"; PED whole page; EIV whole page.

1. I reviewed the sections listed above in the bundle version shown.

2. My comments are in the attached comment table: ____ must-fix, ____ should-fix, ____ suggestions.

3. Result (tick one):

☐ **Passes:** the practical guidance in these sections is accurate and appropriate for a general parent audience, and nothing in them would, in my professional opinion, delay or discourage a family from seeking an evaluation.

☐ **Passes after changes:** as above, once every must-fix item is resolved to my satisfaction in the re-check.

☐ **Does not pass.** Main reason: _____

4. This is editorial review for the publisher. It is not medical advice to any reader and creates no clinician-patient relationship. I did not review how the studies are described, legal wording, or any product.

5. My fee does not depend on my result. My answers to the independence questions are complete and true today. Conflicts to note (including any study on these pages that I wrote): _____

6. Credit (tick one):

☐ No credit.

☐ Named credit on the pages I reviewed (not POL): "Practical guidance on this page reviewed by [Full name], [credential], on [date]. [Name] was paid for this review, did not write this page, and does not endorse any product." Name and credential exactly as they should appear: _____

☐ Unnamed credit: "Practical guidance on this page reviewed by a paid [pediatric speech-language pathologist / developmental-behavioral pediatrician] on [date]."

7. Only if I choose:

☐ You may add my name to the page's structured data (reviewedBy).

8. I understand that my credit appears only on the pages and sections I reviewed, only for this version, never with any product or in advertising, and that I can withdraw it at any time in writing.

Signature: _____ Date: _____

Re-check (within 30 days): version ID _____

☐ All my must-fix items are resolved.

☐ Not resolved: _____

Signature: _____ Date: _____

Clearance form C: claims and legal review

Bundle version ID: _____ Reviewer code: LG-_____ Engagement letter dated: _____

Pages reviewed: IDX, FAQ, GLO, PED, EIV, POL, and the reviewer contract rider.

Checked (tick each one reviewed):

- ☐ Express and implied health claims, including the link to the free play printable and any implied product claim
- ☐ Causation language, disclaimers, and "not medical or legal advice" wording
- ☐ Public promises in the editorial policy and FAQ (conflict of interest, reply times, who checks sources)
- ☐ Descriptions of named researchers, clinicians and organizations (defamation and false light)
- ☐ Quotations, study titles and screening-instrument names
- ☐ Privacy statements (printable, contact form)
- ☐ Reviewer credit wording (FTC Endorsement Guides)
- ☐ Reviewer contract rider and the freelancer template, for Roles A and B

Result (tick one):

- ☐ **Cleared** for publication as written.
- ☐ **Cleared after the changes listed in my memo** (to be confirmed in one follow-up).
- ☐ **Not cleared.** Main reason: _____

This is legal advice to AlphaPlay LLC only. It is not a public endorsement, and it may not be quoted or described publicly without my written consent.

Credit (default none):

- ☐ No credit.
- ☐ Other, exactly as written here: _____

Signature: _____ Date: _____

Follow-up: version ID _____

- ☐ The listed changes are made as advised.

Signature: _____ Date: _____